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Inversa de un morfismo de anillos biyectivo

Ejercicio 1. Sea $f\colon R \to S$ un morfismo de anillos biyectivo, demuestra que $f^{-1}\colon S \to R$ es un morfismo de anillos.

***Solución*:** Sean $a, b \in S$, como f es sobreyectiva, $\exists x, y \in R : f(x) = a \wedge f(y) = b$. Comprobemos las tres propiedades de la `morfismo-anillos/`**??**.

(i) $f^{-1}(a+b) = f^{-1}(f(x)+f(y)) = f^{-1}(f(x+y)) = x+y = f^{-1}(a)+f^{-1}(b)$.

(ii) $f^{-1}(a \cdot b) = f^{-1}(f(x) \cdot f(y)) = f^{-1}(f(x \cdot y)) = x \cdot y = f^{-1}(a) \cdot f^{-1}(b)$.

(iii) Como f es un morfismo de anillos, $f(1_R) = 1_S$. Luego, $f^{-1}(1_S) = f^{-1}(f(1_R)) = 1_R$.

Por tanto, f^{-1} es un morfismo de anillos. ■

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